

Day 2

Architectural and data choices

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When more is better

In many NLP tasks, benchmarks are now saturated for high-resource languages

- POS tagging
- Named Entity Recognition
- Grammatical error correction

In practice, LLMs diminish the need for specialized algorithms

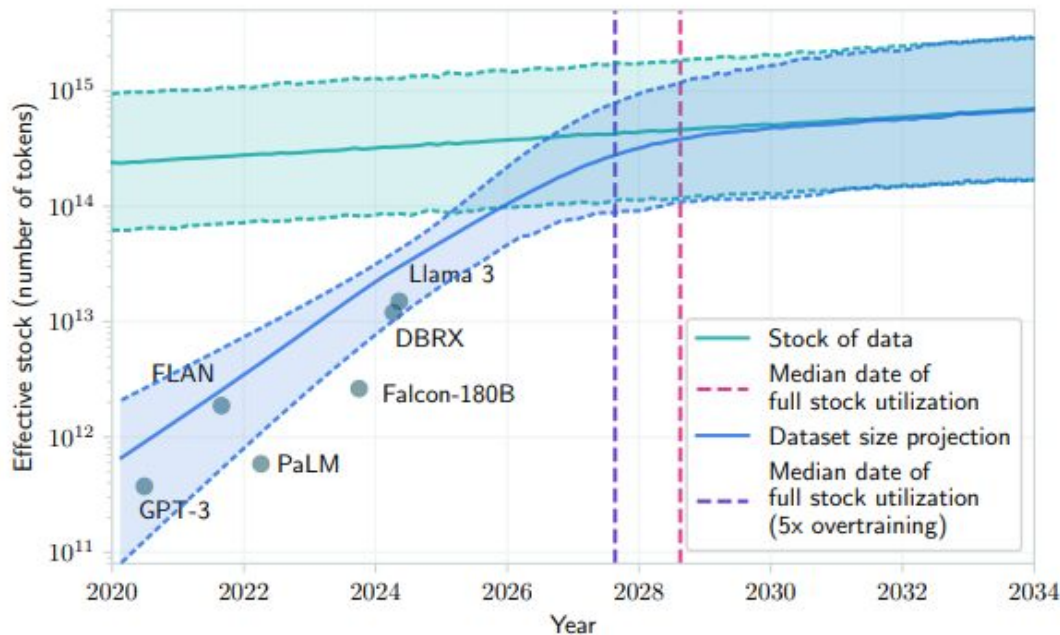
When more is better

In many other tasks, larger models generally perform better than classical methods:

- Machine translation
- Few-shot / in-context learning
- Coding and math reasoning
- Question answering
- Multilinguality

When more is *not* better

Effective stock of human-generated public text and dataset sizes used to train notable LLMs



Villalobos et al. (2024)

When more is *not* better

Data availability and quality issues:

- Most LLM training data sets are proprietary
- Even the available ones are of dubious quality:
 - “The Pile” (Gao et al., 2020)

“we determine that the Pile is 97.4% English. <...> this methodology provides only a rough estimate for English content and no reliable conclusions for low-resource languages can be drawn.”

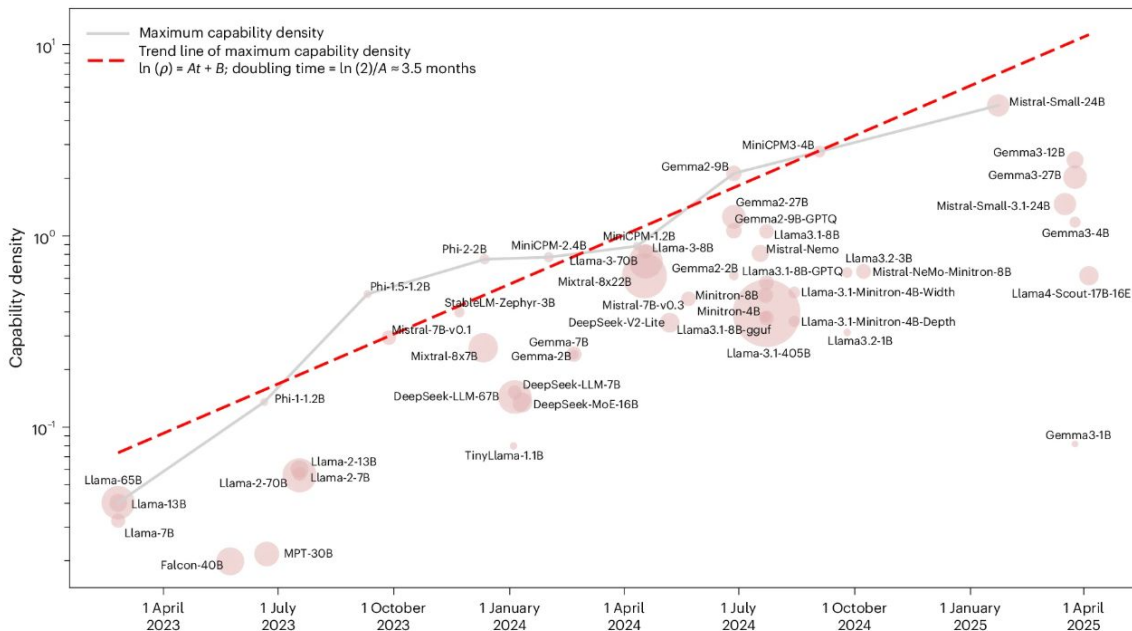


When more is *not* better

Capability density:
capability contained
within each unit of
model parameters

Fig. 1: The estimated capability density of open-source base LLMs.

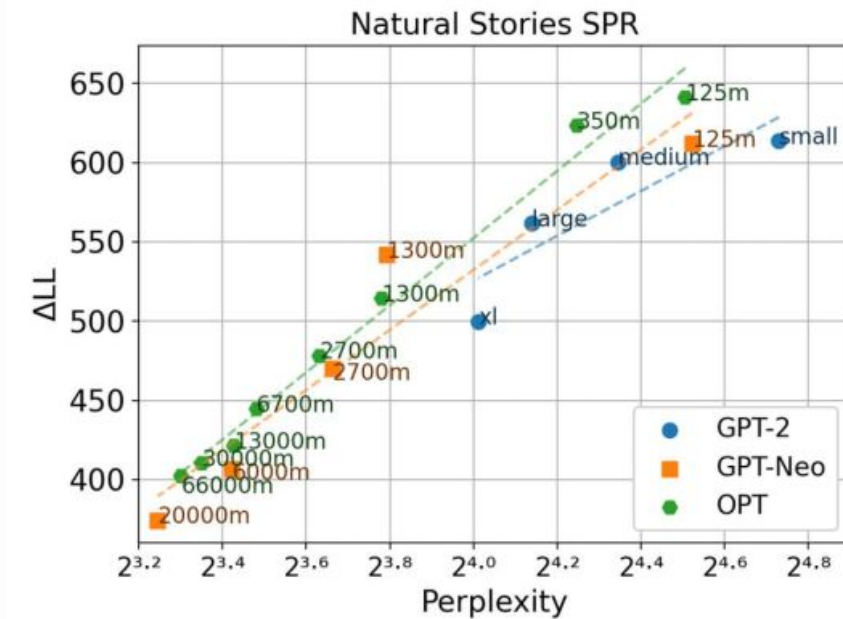
From: [Densifying LLMs](#)



Xiao et al. (2025)

When more is *not* better

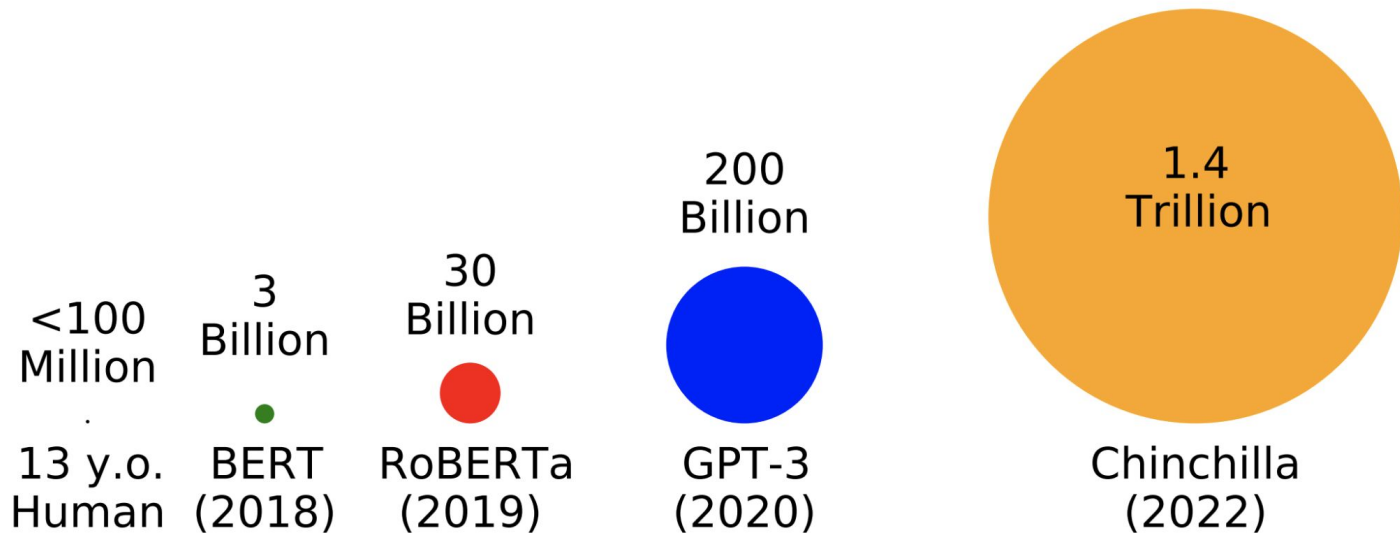
- Larger models have lower perplexity
- But they are associated with worse fit to reading times



Oh & Schuler (2023)

When more is *not* better

BabyLM Challenge: <https://babylm.github.io/>



When more is *not* better

Poverty of the stimulus: the linguistic input children receive is too limited to fully explain the grammatical knowledge they acquire

Piantadosi (2024): *LLMs “essentially lay this issue to rest because they come with none of the constraints that others have insisted are necessary, yet they capture almost all key phenomena”*

vs

Wilcox et al. (2025): *“if the LM is exposed to many thousands of times more sentences than humans, it may encounter plenty of evidence as to the correct generalization that is not available to human learners”*

Issues beyond data

- What does a model bring to the table before it sees the data?
 - Training objective
 - Architecture
 - Input units
- What are the specifics of the learning setup?
 - Amount of training / stopping criteria
 - Optimisation methods
 - Source of the learning signal (raw data / teacher)

Training objective

Autoregressive

predict next token
↑
The → cat → sat → on → the → [mat]

context: left only

Masked

predict masked token
↓
The → cat → [MASK] → on → the → mat
└───┬───┘ └───┬───┘

context: left + right

Which one is more cognitively plausible?

Training objective

- Common-sense wisdom: autoregressive
 - Human comprehension is incremental and predictive
- But: pure forward prediction misses integration
 - Eye movements are not purely left-to-right

Architecture

RNN vs. transformers:

Transformers explain self-paced reading times and neural activity (during reading English sentences) better than RNNs

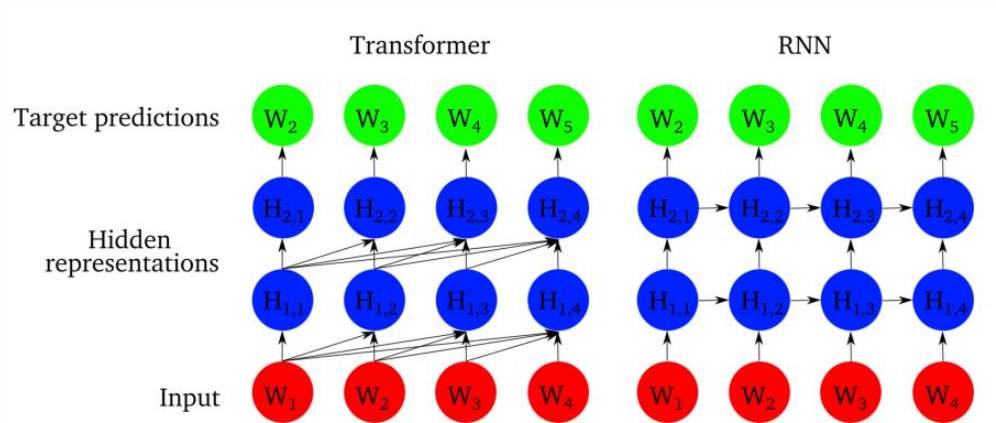


Figure 1: Comparison of sequential information flow through the Transformer and RNN, trained on next-word prediction.

Merkx & Frank (2021)

Architecture

Grammaticality
illusion

The carpenter *who the craftsman* **who the peasant**
carried *hurt* supervised the apprentice in the garden.

V1 V2 V3 D1

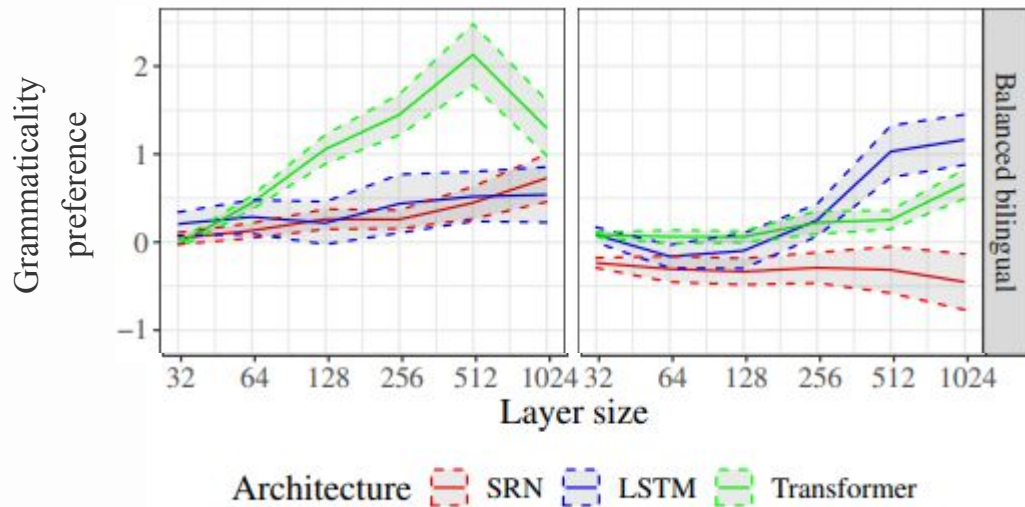
De timmerman *die eergisteren de vakman* **die zater-**
dag de boer droeg *bezeerde* begeleidde de leerling in
de tuin. V1 V2 V3 D1

Roslund & Matushevych (2022)

Architecture

RNN vs. transformers

“All three architectures correctly predict the cognate effect, but only the SRN succeeds at the grammaticality illusion”



Roslund & Matushevych (2022)

Input units

- **Tokenization:** character vs. subword vs. word
- Alternatives: morphological, pixel-based
- Usually treated as a simple preprocessing step
- Can affect psycholinguistic findings

‘If you were to journey’

Finer granularity, more character-like ($|V| = 256$)

▯ I ▯ f ▯ y ▯ o ▯ u ▯ w ▯ e ▯ r ▯ e ▯ t ▯ o ▯ j ▯ o ▯ u ▯ r ▯ n ▯ e ▯ y

Coarser granularity, more word-like ($|V| = 128000$)

▯ If ▯ you ▯ were ▯ to ▯ journey

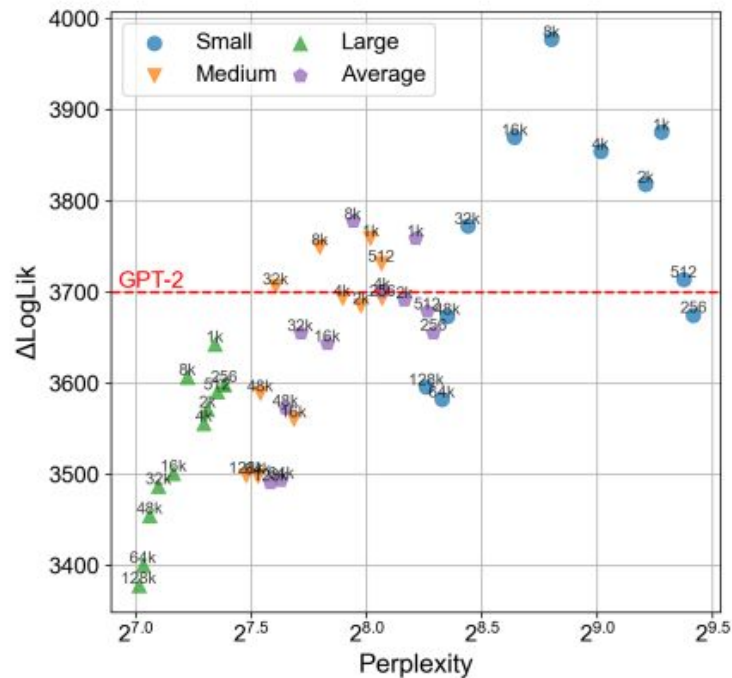
Figure 1: Smaller subword vocabulary sizes result in longer sequences of finer-granularity **tokens** that are more character-like (top), and larger vocabulary sizes result in shorter sequences of coarser-granularity **tokens** that are more word-like (bottom).

Oh & Schuler (2025)

“on gardenpath constructions, LMs trained on coarser-grained tokens generally assigned higher surprisal to critical regions”

Input units

- **Tokenization:** character vs. subword vs. word
- Alternatives: morphological, pixel-based
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Oh & Schuler (2025)

Input units

- Orthography vs. phonology
- Word boundaries

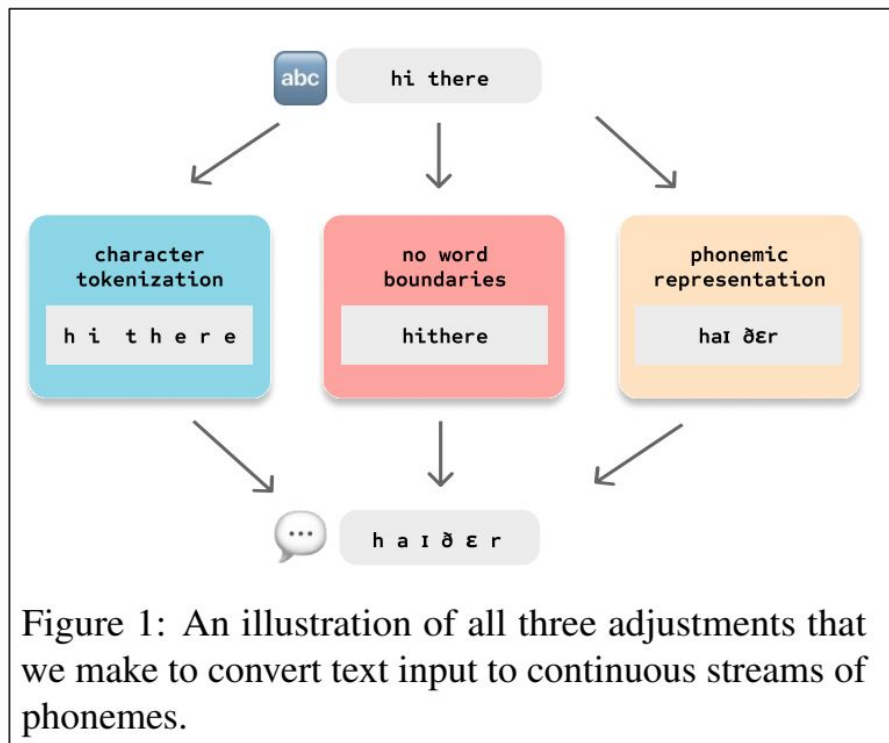


Figure 1: An illustration of all three adjustments that we make to convert text input to continuous streams of phonemes.

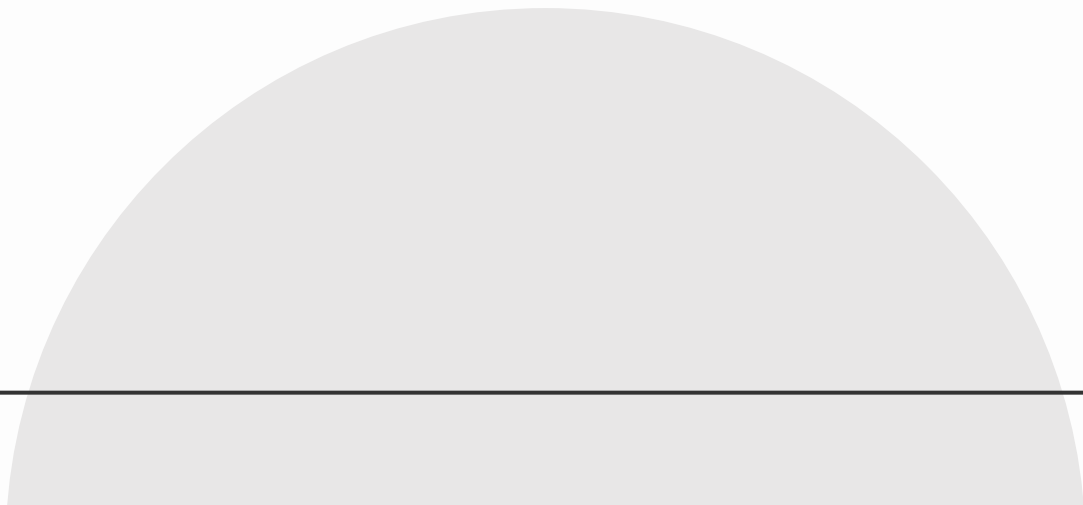
Goriely et al. (2024)

Specifics of the learning setup

For higher cognitive plausibility:

- Number of epochs should not be unreasonably high
- Stopping criteria need to be set in a principled way
- Choice of checkpoints to use need to be principled

**Making models more
data-efficient**



Architectural improvements

- LTG-BERT (Samuel et al., 2023) – re-engineered BERT internals (e.g., disentangled position encoding)
- ELC-BERT (2023 winner, Charpentier & Samuel, 2023) – layers can reach back to any earlier layer ("every layer counts")
- GPT-BERT (2024 winner, Charpentier & Samuel, 2024) – one model trained using both an autoregressive and masked objective

Data ordering / selection

- Curriculum learning: teach the model like a child, moving from simple to complex
- Arrange data by length, syntactic complexity, frequency, domain (child-directed-first)
- Largely unsuccessful in BabyLM, only modest improvements

Data ordering / selection

- An alternative that worked: select data by the model's own signal (not human difficulty)
 - Schoenegger et al. (2025): prioritize more prototypical examples
 - Fysikoudi et al. (2025): training on inputs the model finds most uncertain
- Other methods: text simplification, synthetic data augmentation

Knowledge distillation

- Idea: train a small “student” to imitate a “teacher” model's output distribution (its soft probabilities), not just the ground-truth next word
- Timiryasov & Tastet (2023): BabyLlama, a small model trained by two larger teacher models
- 2025 - Interaction track in BabyLM, the teacher can be a large model (beyond the allocated data budget)

Multimodal grounding

- A separate track in 2024 and 2025 BabyLM
- In 2024, no submissions outperformed the baselines
 - Under a tight data budget, adding vision tends to hurt language-only performance
- In 2025, some progress
 - Ganescu et al. (2025): lightweight dynamic gating - learned to use vision for content words, text for function words

“Boring” solutions

At this scale of training (10M-100M tokens), small design choices can give meaningful gains:

- tokenization
- sequence length
- optimizer and batch settings

BabyLM - summary of approaches

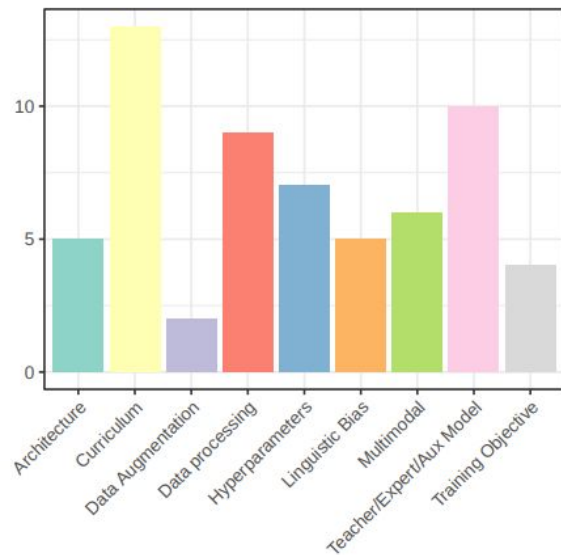
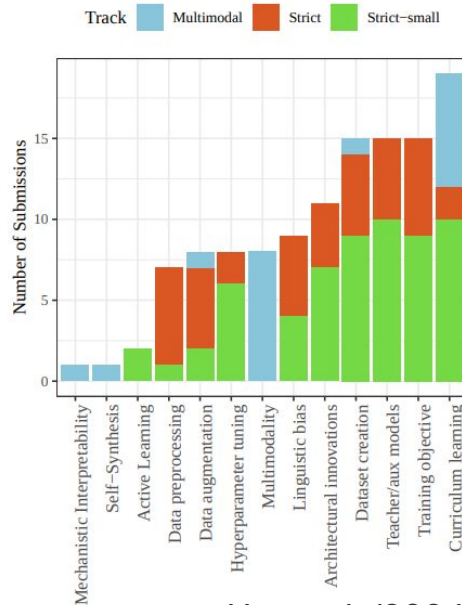


Figure 3: Total number of submitted models that used each of the nine approaches in our typology. We count at most one submitted model per participant per track.

Warstadt et al. (2023)



Hu et al. (2024)

Figure 1: A breakdown of the various approaches used in the 2024 BabyLM challenge, organized by category and track. Curriculum learning again takes the top spot as the most popular approach, followed by training objective innovations.

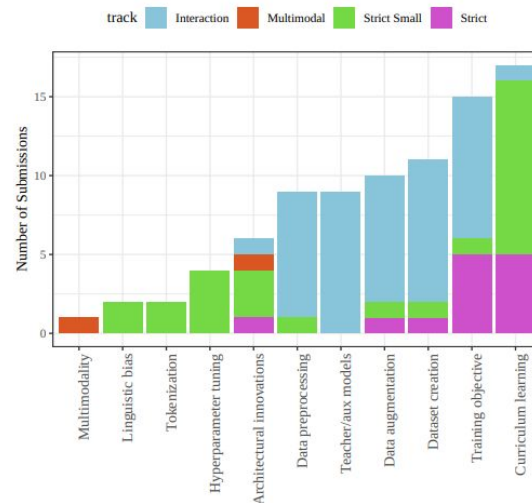


Figure 1: Number of submissions by approach. Curriculum learning was again the most popular approach. This year, we encouraged more teacher/auxiliary-model approaches in the interaction track.

Charpentier et al. (2025)

Data issues again

- One assumption behind BabyLM: developmentally plausible input must be helpful
- In child language acquisition: CDL facilitates learning
- But Kempe et al. (2024) argue that the evidence for such facilitation mostly comes from specific domains (e.g., prosody)
- So does CDL help learning in LMs?
 - Evaluations are often syntactic

Data issues again: CDL and syntax learning

Padovani et al. (2025):

- Train LMs on either CDL or Wikipedia
 - GPT-2 and RoBERTa
 - English, French, German
- Test them on syntactic benchmarks:
 - BLiMP, Zorro, CLAMS

Data issues again: CDL and syntax learning

Padovani et al. (2025):

- Mixed results overall
- On the most suitable benchmark CLAMS, after controlling for frequency, the Wikipedia models are better

MIXED RESULTS! —————→ CHILDES $\approx$ Wiki CHILDES $>$ Wiki CHILDES $<$ Wiki

	Model	Training Data	BLiMP	Zorro	CLAMS		
					English	French	German
GPT-2	CLM	CHILDES	0.61 ± 0.02	0.76 ± 0.04	0.60 ± 0.01	0.64 ± 0.01	0.69 ± 0.03
		Wiki	0.61 ± 0.02	0.69 ± 0.04	0.71 ± 0.01	0.80 ± 0.01	0.81 ± 0.01
RoBERTa	MLM	CHILDES	0.59 ± 0.03	0.66 ± 0.05	0.57 ± 0.02	0.59 ± 0.02	0.70 ± 0.01
		Wiki	0.59 ± 0.03	0.67 ± 0.03	0.63 ± 0.01	0.69 ± 0.01	0.75 ± 0.01

Data issues again: CDL and syntax learning

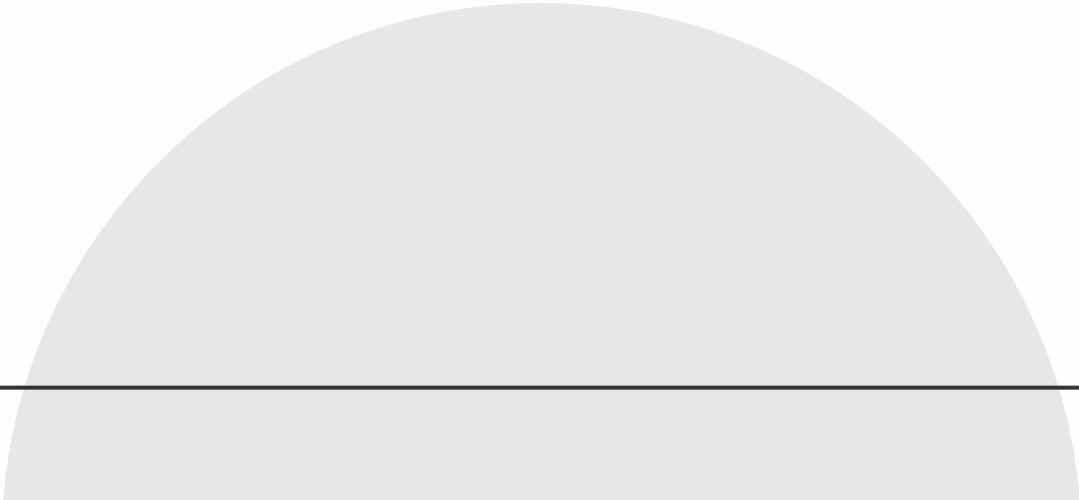
Padovani et al. (2025):

- Mixed results overall
- On the most suitable benchmark CLAMS, after controlling for frequency, the Wikipedia models still show an advantage

Conclusion:

- CDL doesn't seem to help syntax learning in LMs
- Perhaps we need a different kind of models - interaction, communication, pragmatic context?

Making modeling decisions



What are we trying to explain?

- Acquisition - how the child gets to competence from limited input
- Competence - what the (adult) grammar looks like
- Processing - real-time comprehension in (adult) speakers

What counts as explaining?

- Greater model power does not by itself make a stronger theory
- A system that can predict many behaviors explains little unless we can say why it predicts them and how its mechanisms map onto the human system
- In some LLM work, the model has become the proximate object of study

What counts as explaining?

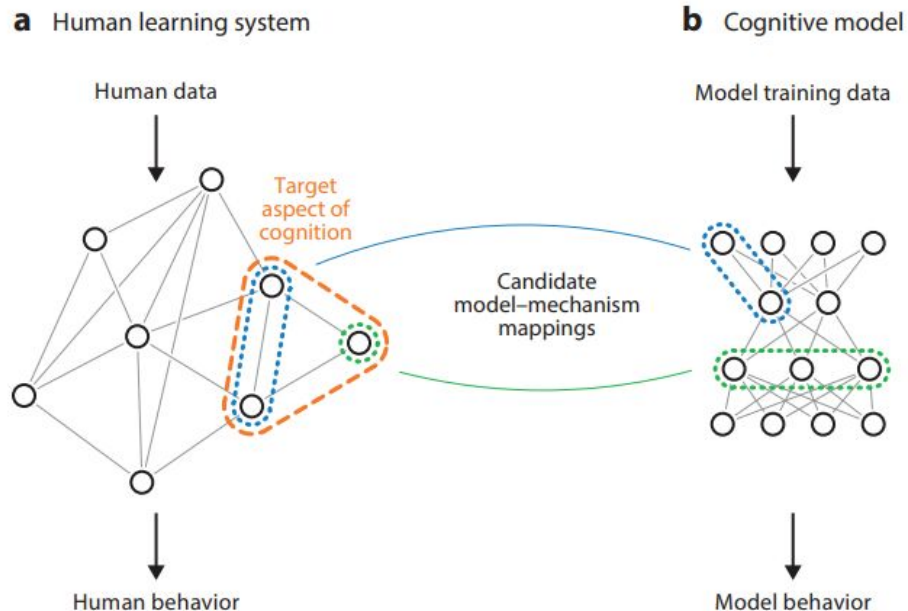


Figure 1

A schematic depiction of a model-mechanism mapping between (a) a human learning system and (b) a cognitive model. Candidate model-mechanism mappings are pictured as mapping between representations but can also be in terms of input data, architecture, or learning objective.

LLM surprisal and reading times

LLM surprisal and reading times

$$\text{surprisal}(w) = -\log_2 P(w \mid \text{context})$$

Linking claim: reading time rises roughly linearly with surprisal
(Smith & Levy, 2013)

The same LM produces both a perplexity value (how good of a model it is) and word-by-word surprisal values (its prediction about human reading effort)

LLM surprisal and reading times

Approach:

- Fit a regression: reading time $\sim$ baseline predictors (word length, frequency, position)
- Add surprisal and ask how much the model's log-likelihood improves
- ΔLL is what surprisal explains over and above the other predictors

LLM surprisal and reading times

- Size (Oh & Schuler, 2023, *TACL*): across model sizes, lower perplexity is associated with worse reading-time fit.
- Training amount (Oh & Schuler, 2023, *EMNLP*): fit peaks at $\sim 2\text{B}$ training tokens ($\approx$ step 1000 for Pythia), then declines.
- Mechanism (Oh et al., 2024) - frequency: bigger (over-trained) models predict rare words too easily and *underpredict* the human slowdown on them.

LLM surprisal and reading times

- Pythia (Biderman et al., 2023): same data in the same order across sizes, 154 released checkpoints
- Easy to manipulate *training step* and *model size* keeping the rest fixed

LLM surprisal and reading times

The Natural Stories Corpus (Futrell et al., 2021):

- 181 participants
 - read 10 naturalistic English stories
 - a total of 10,245 tokens
-
- You can average by-word times over many readers

LLM surprisal and reading times

- Goal: reproduce the qualitative pattern on a subset of one corpus with a simplified regression model
- A demo, not a full replication:
 - Only 1 data set, and a subset of that
 - OLS instead of mixed-effects models

Thanks!

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